IMAGES OF NOETHERIAN SPACES UNDER CONTINUOUS MAPS ARE NOETHERIAN, AND IMAGES OF LOCALLY NOETHERIAN SPACES UNDER OPEN MAPS ARE LOCALLY NOETHERIAN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Noetherian and locally Noetherian properties, and the correspondence between closed subsets of the image and preimages of closed subsets under a continuous map. Set up in the author's section on Noetherian topological spaces, immediately before the counted statement. Fixes the two assertions and the mechanism by which chains in the image are lifted to the source.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces.*

(1) *If X is Noetherian, then $f(X)$ is Noetherian.*
(2) *If X is locally Noetherian and f open, then $f(X)$ is locally Noetherian.*

Proof. In case (1), suppose that $Z_1 \supset Z_2 \supset Z_3 \supset \ldots$ is a descending chain of closed subsets of $f(X)$ (as usual with the induced topology as a subset of Y). Then $f^{-1}(Z_1) \supset f^{-1}(Z_2) \supset f^{-1}(Z_3) \supset \ldots$ is a descending chain of closed subsets of X. Hence this chain stabilizes. Since $f(f^{-1}(Z_i)) = Z_i$ we conclude that $Z_1 \supset Z_2 \supset Z_3 \supset \ldots$ stabilizes also. In case (2), let $y \in f(X)$. Choose $x \in X$ with $f(x) = y$. By assumption there exists a neighbourhood $E \subset X$ of x which is Noetherian. Then $f(E) \subset f(X)$ is a neighbourhood which is Noetherian by part (1). □

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